

**Task-Initiation Failure as an Underidentified Endpoint: A Framework for Distinguishing
Sources of the Goal-to-Action Gap**

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Abstract

Failure to begin an intended task is often interpreted as evidence of weak motivation, insufficient intention, or poor executive control. For motivation science, however, non-initiation is an underidentified endpoint: a person can endorse a superordinate goal ($G = 1$) yet fail to cross an objective initiation criterion ($Y^I = 0$) for functionally different reasons. This conceptual article proposes a candidate identification framework for the goal-to-action transition. It distinguishes action construction (F1), local target-policy selection (F2), a self-generated-versus-externally-supported onset contrast (F3), and transition-sensitive initiation cost (F4), while treating downstream execution failure as a separate endpoint. The framework specifies ordered perturbations, proximal measures where noncircular assays are available, explicit conditioning on local policy selection before onset contrasts, and prospective rules for mediation, nesting, merger, dropping contrasts, invalid tests, and lower-dimensional alternatives. F3 remains operational because no independent proximal assay is currently established, and F4 remains a broad transition-sensitive family. The proposal builds on established intention-behavior, Action Control Theory, implementation-intention, control-allocation, and target-engagement literatures; its residual contribution is an ambiguity-reduction architecture for task initiation. It offers a falsifiable conceptual program rather than a validated biological factorization.

Keywords: task initiation; motivation; action control; intention-behavior gap; causal identification

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Introduction

A person may know what needs to be done, endorse doing it, understand the consequences of delay, possess the necessary ability, and nevertheless fail to begin. Such episodes are described using terms including procrastination, executive dysfunction, motivational impairment, avoidance, poor self-regulation, and impaired initiation. These labels can be useful descriptively, but they do not by themselves identify the causal process that generated the behavior.

Consider the observation

$$G = 1, \quad Y^I = 0,$$

where $G = 1$ denotes representation and endorsement of a superordinate goal at the start of the bounded episode, not selection of the local target policy. $Y^I = 0$ denotes failure to cross a prespecified task-initiation criterion. The criterion must be observable and declared in advance for the task at hand - for example, the first valid task-directed action or another objective onset marker. From this observation alone, the inference

$$G = 1, \quad Y^I = 0 \Rightarrow \text{weak motivation}$$

does not follow. Nor does the same endpoint uniquely imply deficient executive capacity, forgetting, excessive inhibition, low reward sensitivity, or any other single latent cause.

The problem is central to motivation science because the transition from wanting or intending to acting is already recognized as a multi-step achievement rather than a guaranteed consequence of desire. The intention-behavior literature documents systematic failures of

enactment (Sheeran & Webb, 2016). Kruglanski (2017), writing in *Motivation Science*, describes motivational phases in which a goal must prevail over competing concerns, effective means must be identified, and sufficient energetic resources must be available before goal-directed action occurs. Implementation-intention research shows that desired outcomes can remain insufficiently linked to executable responses (Gollwitzer, 1999), while control-allocation models emphasize value, efficacy, and cost (Shenhav et al., 2013). Action-control research likewise distinguishes maintained intentions from efficient enactment, and the modern Action Control Theory synthesis emphasizes the coordination of intention memory, affect regulation, and behavior control (Kazén et al., 2008; Koole et al., 2023). The present article does not claim these component distinctions or the general intention-to-action problem as new.

The narrower question is one of attribution at the task-initiation boundary: given the common endpoint $G = 1$ and $Y^I = 0$, what candidate functional distinctions are worth testing, what ordered contrasts reduce ambiguity among them, and what evidence should force the proposed decomposition to contract? The candidate organization contains four contrasts—action construction (F1), local policy selection (F2), self-generated versus externally supported onset (F3), and transition-sensitive initiation cost (F4). Post-initiation execution is retained only as an exclusion contrast so that failure after onset is not misclassified as failure to initiate.

The contribution is therefore not a new theory of motivation, a rediscovery of the intention-behavior gap, or a replacement for Action Control Theory. It is a task-initiation-specific identification architecture: strict non-initiation is treated as an underidentified endpoint; perturbations are ordered to reduce specific ambiguities; proximal measures are required where noncircular assays are available; conditioning variables are made explicit; and prospective rules

allow candidate contrasts to be nested, merged, dropped, or replaced by lower-dimensional alternatives. Added dimensionality is an empirical burden rather than an assumption.

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Operational Boundary and Episode Variables

Boundary Declaration

The target of analysis is a bounded goal-to-action episode. The focal person is the behaving organism whose transition from an endorsed superordinate goal to an observable task-directed onset is analyzed. The controlled task environment comprises task structure and manipulated variables that can alter that episode, including action prompts, reward or efficacy contingencies, initiation cues, and repeat-versus-switch requirements. The framework does not assume that every determinant is internal to the person; it requires the analysis boundary, observables, conditioning variables, and manipulated inputs to be declared so that claims about dependence are interpretable.

Let x_t denote person state at time t , e_t the experimentally controlled task environment, and θ_i history-bearing parameters for individual i . Let G denote superordinate goal

representation/endorsement, A a proximal measure of local action specification or construction, P local target-policy selection, Y^I whether the prespecified initiation criterion is crossed, T^I latency to that criterion when onset occurs, and Z^E a downstream execution outcome after onset. The baseline equivalence class of interest conditions on $G = 1$ and $Y^I = 0$; it does not condition on $P = 1$.

Operationally, $G = 1$ should be established independently of the local choice that P represents. For example, before action or policy manipulations, a participant could identify the target outcome as currently intended or endorsed and accurately state that outcome. That establishes representation/endorsement of the superordinate goal for the bounded episode; it does not establish that the target action is specified, that the target policy currently wins, or that onset will occur.

$$(P_{i,t}, Y^I_{i,t}, T^I_{i,t}) = f(F_1, F_2, F_3, F_4, x_t, e_t, \theta_i).$$

F_1 - F_4 are candidate functional contrasts within a hypothesis class, not demonstrated anatomical modules, orthogonal latent factors, or validated causal factors. Their implementations may overlap, recur, or interact. A downstream execution outcome Z^E is conceptually separate: it is measured only after $Y^I = 1$ and is not an alternative realization of baseline non-initiation.

$$Z^E_{i,t} = g(Y^I = 1, d_t, x_t, e_t, \theta_i),$$

where d_t denotes post-initiation execution demand. This separation fixes the scope condition: an action that has already crossed the initiation criterion cannot simultaneously count as another realization of $Y^I = 0$.

Hierarchy of Episode Variables and Conditioning Sets

The hierarchy is a conditioning structure for analysis, not a claim of serial biological architecture. F1 and F2 are evaluated before or during establishment of local target-policy selection. F3 and F4 are onset contrasts evaluated only when $P = 1$ has been observed or made sufficiently stable for the task and an executable action is available. Thus F2 can generate $G = 1$, $Y^I = 0$ through failure of the target policy to win, whereas F3 and F4 concern the narrower subset $G = 1$, $P = 1$, $Y^I = 0$. The baseline endpoint remains the same; the conditioning set becomes stricter as downstream onset contrasts are tested.

Conditioning later onset contrasts on $P = 1$ also creates a measurement problem: querying, forcing, or repeatedly confirming a local choice can itself change confidence, commitment, preparation, or the onset state subsequently tested. An empirical implementation should therefore use a sufficiently nonreactive measure where possible, model or estimate measurement reactivity, separate the P assessment from the later onset test, or otherwise show that establishing P did not manufacture the F3 or F4 effect.

Observational Equivalence and Candidate Functional Organization

Suppose two candidate models M_a and M_b both yield $Y^I = 0$ under the declared baseline condition $G = 1$. They are observationally equivalent with respect to that endpoint even if they posit different internal functional organizations. Baseline non-initiation therefore defines an equivalence class, not a unique mechanism.

A perturbation family becomes informative when candidate models make discriminating predictions under admissible interventions:

$$P(Y^I | I, M_a) \neq P(Y^I | I, M_b).$$

This predictive divergence is necessary but not sufficient for stronger causal interpretation. A manipulation may affect arousal, expectancy, perceived efficacy, conflict, preparation, attention, or learning in addition to its intended variable. Accordingly, factorization sensitivity here means only that the baseline endpoint admits candidate internal organizations that can, in principle, be separated by informative perturbations. Stronger attribution requires valid implementation, proximal evidence where a non-circular assay is available, convergent discrimination against neighboring or common-factor accounts, and superior held-out prediction relative to lower-dimensional alternatives.

What Baseline Observation Cannot Establish

Even a rich baseline description - the goal is endorsed, the task is understood, relevant ability has been demonstrated elsewhere, and initiation still fails - is insufficient to distinguish at least four initiation-relevant possibilities:

- no sufficiently concrete next action has been constructed;
 - the local target policy fails to win against competing policies;
 - the target policy has been selected ($P = 1$) but self-generated onset remains delayed or absent;
- or
- entry into the target task configuration carries a transition-sensitive initiation cost.

A fifth observation must be kept separate: the person may initiate normally and then fail during execution. Such a downstream failure is scientifically relevant but is not another factorization of $Y^I = 0$. Measuring it separately protects the initiation analysis from endpoint misclassification.

Worked Equivalence Case: Same Endpoint, Different Next Inference

Consider two hypothetical episodes with the same baseline observation: $G = 1$ and $Y^I = 0$. In Episode A, the person endorses the target goal but cannot yet specify a sufficiently determinate next action; a prespecified A measure is correspondingly weak or uncertain. In Episode B, an executable action is already available and A is adequate, but a pre-onset choice measure shows that the target policy does not currently win ($P = 0$). The outward endpoint is identical, but the next informative test is not. Episode A calls for an action-specification test with concurrent P measurement; Episode B calls for a policy-selection manipulation or measurement before any onset contrast is interpreted.

Nothing about $Y^I = 0$ alone licenses either attribution, and these two cases do not exhaust the equivalence class. Their purpose is to show what the architecture changes: the same failure endpoint can require different next observations, and the admissible explanation becomes narrower only after those observations are obtained.

Candidate Contrasts in the Goal-to-Action Transition

The component distinctions below motivate candidate functional contrasts rather than validated mechanisms. Their scientific value in this framework depends on whether future studies can establish manipulation validity, obtain independent proximal evidence where such evidence is available, and show discriminating perturbation-response patterns that outperform lower-dimensional alternatives.

F1—Action Construction: Represented Outcome Versus Executable Next Action

A goal can specify an outcome without specifying an immediately executable response. The intention “finish the project” contains less local action information than a cue-linked plan

such as “when condition X occurs, perform action A.” Implementation intentions can improve goal enactment by linking anticipated situations to goal-directed responses (Gollwitzer, 1999).

At the functional level, the candidate distinction is

$$F_1: G_abstract \rightarrow a_local,$$

where $G_abstract$ is the represented outcome and a_local is an action executable in the current state. Failure to construct a_local is not equivalent to low outcome value. In an identification study, supplying a_local is only a proposed F1-sensitive perturbation because the same intervention can increase perceived efficacy or target-policy advantage. F1 therefore requires a proximal action-construction measure A together with concurrent measurement of F2-relevant quantities; action supply alone cannot identify F1.

F2—Local Policy Selection: Target Value Is Relative to Alternatives

Once candidate actions exist, they compete. Expected-value-of-control theory treats control allocation as depending on expected benefits, efficacy, and costs rather than on abstract outcome importance alone (Shenhav et al., 2013). For the present framework, the essential point is comparative rather than neuroanatomical.

$$\Delta Q = Q(\pi_target) - Q(\pi_competitor).$$

The competitor may be another task, avoidance, waiting, continuation of the current activity, or an alternative solution. High long-term importance therefore does not entail that the target policy currently wins. Opportunity-cost work provides an example: reward foregone by remaining inactive influences free-operant self-initiation latency (Nair et al., 2023). The candidate transformation is

$$F_2: \{\pi_1, \dots, \pi_n\} \rightarrow \pi^*.$$

An F2-sensitive test requires evidence that the manipulation changed local target-policy preference or choice propensity P before any claim is made about its contribution to initiation. Because F1 manipulations can alter efficacy, F2 measurement must also be used when adjudicating whether an apparent action-construction effect is mediated through policy selection.

F3—Self-Generated Versus Externally Supported Onset: An Operational Contrast

Local target-policy selection is still insufficient for onset. Kazén et al. (2008) showed that intentions can remain represented while self-initiation is disproportionately delayed under demanding conditions and that external cues can reduce this disadvantage. The present framework uses this literature to motivate an operational contrast between self-generated onset and externally supported onset after $P = 1$ has been observed or made sufficiently stable.

Neurological evidence demonstrates an extreme dissociation without implying equivalence to ordinary procrastination or ADHD. Schmidt et al. (2008) reported patients with basal-ganglia damage who could register incentive value and produce instructed force while showing abnormal self-generated incentive-related activation. Such evidence supports the logical separability of valuation, instructed capacity, and self-generated activation; it does not provide an independent proximal assay of F3 in the present task family.

$F_3: P = 1; \text{self-generated onset} \leftrightarrow \text{externally supported onset},$

where the bidirectional symbol denotes an experimental comparison rather than a biological transformation. External cue delivery and detection, matched alerting, preparation controls, and convergent forms of onset support can establish manipulation validity and reduce nuisance explanations. They do not independently demonstrate target engagement of a latent 'enactment coupling' mechanism. Until a non-circular proximal assay exists, F3 remains an

operational self-generated-versus-supported onset contrast and its stronger mechanistic interpretation is OPEN.

F4—Transition-Sensitive Initiation Cost

Even when an action is selected and externally cued, initiating it can require changing the currently active task configuration. Task-switching research repeatedly demonstrates slower and often less accurate performance on switch than repeat trials, with preparation reducing but not necessarily eliminating switch costs (Monsell, 2003).

The present framework deliberately treats F4 as a transition-sensitive operational contrast, not as a uniquely localized “control-state transition mechanism.” A repeat-versus-switch effect can reflect reconfiguration, interference, inhibition, stimulus-response remapping, working-memory updating, preparation, or combinations of these processes. A residual switch cost under matched cueing therefore supports sensitivity to a transition requirement; narrower causal attribution requires converging manipulations or assays.

If established switching, preparation, conflict, remapping, or related task-control constructs fully account for the repeat-versus-switch effect, the analysis should contract to that lower-dimensional explanation. A residual behavioral switch effect does not by itself justify an independently necessary F4 mechanism; F4 is retained only as the broad operational transition contrast that the evidence still requires.

This operational distinction remains useful because difficulty entering a task can coexist with strong persistence once the task configuration is active. Goal shielding further supports the broader point that entry and continuation should not be inferred from one another (Shah et al., 2002).

Post-Initiation Execution as an Exclusion Contrast

Selecting and initiating an action does not guarantee successful production of the control or effort required after onset. The Choice-Execution model distinguishes prospective choice from effort production during execution (Ludwiczak et al., 2020), and Winter et al. (2019) reported similar high-effort choice rates in children with ADHD and controls but lower successful execution in the ADHD group.

These findings justify a downstream measurement safeguard: post-initiation execution failure should be measured so that it is not misclassified as task-initiation failure. This contrast is not numbered as an initiation function, does not enter the F1-F4 factorization, and cannot explain $Y^I = 0$ on a trial where Y^I has already equaled 1.

Positioning Within Motivation and Action-Control Science

The framework is intentionally positioned inside an established literature on the movement from goals and intentions to action. Sheeran and Webb (2016) synthesize the intention-behavior gap; Kruglanski (2017) describes motivational phases on the road to action, including goal predominance, means identification, and energetic resources; and Koole et al. (2023) provide an integrative review of Action Control Theory focused on individual differences in intention enactment. These traditions establish that goal endorsement is not equivalent to successful action initiation. They also constrain the novelty claim: the present article does not introduce the general goal-to-action problem or the broad processes that contribute to it.

The identification standard also has close methodological precedents. The NIH Science of Behavior Change experimental-medicine approach emphasizes specifying a candidate target, measuring that target, perturbing it, establishing target engagement, and testing whether engagement changes behavior (Nielsen et al., 2018; Riddle & Science of Behavior Change

Working Group, 2015). Recent behavioral-measurement work similarly frames controlled perturbation as a system-identification problem in which competing models are evaluated by informative perturbations and generalization to unseen perturbations (Cipresso, 2026). These precedents directly support the principle that endpoint change alone does not identify the process responsible for the endpoint.

The residual contribution is narrower and task-initiation specific: (a) a strict initiation endpoint treated as an equivalence class; (b) a candidate F1–F4 organization plus an explicitly downstream execution exclusion; (c) an ordered ambiguity-reduction protocol, including direct F1/F2 adjudication and explicit conditioning on P before onset contrasts; and (d) prospective rules that allow evidence to support mediation or nesting, merge or drop contrasts, favor a common factor, reject a stable person-level profile, or declare a proposed selective test invalid. The scientific burden is whether this organization improves discrimination and held-out prediction beyond the lower-dimensional alternatives with which it competes.

Candidate Functional Decomposition and Evidence Levels

Table 1 summarizes the candidate functional factorization for strict initiation. The four rows are not homogeneous claims: F1 and F2 are paired with proposed proximal variables, F3 is currently an operational onset contrast without an independent non-circular proximal assay, and F4 is a broad transition-sensitive family. The table therefore defines a hypothesis class and adjudication scheme, not four validated latent mechanisms.

Table 1*Candidate functional contrasts for strict task initiation*

Contrast	Candidate discriminating perturbation	Evidence / conditioning before stronger attribution	Contraction / alternative outcome
F1 Action construction	Supply an explicit state-action mapping versus self-generation while holding payoff/cueing approximately constant.	Measure proximal action-specification variable A separately from onset; concurrently measure efficacy and local policy preference P because specification can change F2.	Support separability only if A-specific evidence remains after F2 mediation tests and improves held-out prediction; otherwise nest/merge F1 with F2. Invalid if A is not engaged.
F2 Local policy selection	Manipulate relative payoff, efficacy, opportunity cost, or competitor availability before local choice.	Measure target-policy preference/choice propensity P before onset; monitor A so F1-mediated changes are not silently relabeled F2.	Nest/merge/drop F2 if P adds no held-out predictive information beyond F1 or a lower-dimensional/common-factor model.
F3 Operational onset support	Conditional on P = 1 and an available action, compare self-paced onset with externally supported onset using matched alerting/preparation controls.	Cue delivery/detection and nuisance-control success establish manipulation validity; Y ¹ /T ¹ are endpoint measures. No independent non-circular proximal F3 assay is currently claimed.	Drop/merge F3 if no reproducible residual self-generated-versus-supported onset contrast survives valid convergent tests; mechanistic interpretation remains OPEN.
F4 Transition-sensitive initiation cost	With onset support approximately matched, compare equally trained repeat versus switch requirements.	Verify repeat/switch assignment, familiarity, and cueing; this is manipulation validity for a broad operational contrast, not target engagement of a unique mechanism.	Drop F4 if transition-sensitive costs disappear under adequate tests; retain only broad transition sensitivity unless converging evidence supports narrower structure.

Functional Contrasts Are Not Serial Biological Stages

F1-F4 should not be read as a mandatory temporal pipeline. Action construction can alter expected efficacy; policy evaluation can recur; external cueing can alter preparation; and transition demands can change which actions are available. The labels are experimentally useful only to the extent that converging evidence and held-out prediction support retaining them as separate terms.

More Than One Contrast May Constrain the Same Episode

A poorly specified task may also have a low relative policy advantage, and an external cue may affect both alerting and onset support. The target is therefore not one hidden cause per episode. The target is the smallest candidate decomposition justified by valid perturbations, proximal evidence where available, nuisance control, and out-of-sample model comparison.

Identification Standard: From Perturbation to Attribution

Differential intervention response can be informative but is not, by itself, function identification. The evidential chain must distinguish implementation from proximal engagement and endpoint response. A general sequence is:

manipulation assignment → manipulation-validity evidence → independent proximal evidence
 where available → initiation response → convergent discrimination → held-out model
 comparison.

Only after the relevant parts of this chain are supported should stronger causal language be considered. A valid manipulation can exist without a valid proximal target assay, as is presently the case for F3. Before stronger evidence is available, terms such as 'candidate functional contrast,' 'operational contrast,' 'proposed discriminating perturbation,' or 'evidence consistent with selective engagement' are more accurate.

Manipulation Validity, Proximal Evidence, and Endpoint Response

Manipulation validity asks whether the assigned input was delivered, perceived, and implemented as designed. Independent proximal evidence asks whether a prespecified variable distinct from the initiation endpoint changed. For F1, a separate action-specification measure A can test whether the local action became more determinate. For F2, target-policy preference or

choice propensity P can test whether local policy advantage changed. F3 presently lacks a defensible non-circular proximal assay: cue delivery/detection and matched alerting establish manipulation validity and control nuisances, while Y^I/T^I remain endpoint measures. F4 likewise uses verified repeat-versus-switch assignment as manipulation validity, not as proof of a unique transition mechanism.

A manipulation check is not automatically a mechanism assay. The strongest designs distinguish (a) whether the assigned manipulation was implemented, (b) whether an independently defined proximal variable changed where such a variable exists, (c) whether the strict initiation endpoint changed, and (d) whether neighboring or lower-dimensional accounts predict the same held-out responses.

Progressive Ambiguity Reduction, Not Latent-State Clamping

The ordered design is best understood as progressive ambiguity reduction rather than latent-state clamping, because fixing an observable input does not guarantee that a latent process has been fixed. Supplying an action can still alter efficacy; fixing a displayed choice does not eliminate value or confidence effects; an external cue does not eliminate preparation; and repeated post-initiation demand can generate expectancies that influence later trials.

The ordering nevertheless has value. Before interpreting the operational F3 onset contrast, action specification can be measured and local target-policy selection P can be observed or made as stable as feasible. Before interpreting F4 transition sensitivity, onset support can reduce one source of self-generated onset variability. The logic is experimental ordering of constraints, not a serial biological architecture and not a guarantee that latent states are fixed.

Future implementations should therefore include counterbalancing, randomization, practice stabilization, order controls, and where relevant trial structures that prevent information

about later demand from leaking into earlier decisions. Repeated exposure should be modeled or controlled when it can create learning or expectancy effects.

Invalid Selective Test Versus Negative Evidence

A failed dissociation has distinct interpretations. If the assigned manipulation is not validly implemented, or if a prespecified proximal variable such as A or P fails to change when that change is required by the test, the selective test is invalid and cannot support factor collapse. If the test is valid, the relevant proximal evidence is obtained where available, and the predicted distinct initiation pattern still fails across converging implementations, the result can support merger, dropping a contrast, or a lower-dimensional model. For F3 and F4, where no unique proximal mechanism assay is claimed, validity and nuisance-control criteria must be prespecified so that 'insufficient selectivity' cannot be invoked post hoc to protect the contrast indefinitely.

This distinction prevents imperfect selectivity from becoming an unfalsifiable rescue. Empirical protocols should prespecify proximal engagement thresholds where a distinct proximal assay exists, manipulation-validity criteria where it does not, expected nuisance checks, and the model comparisons that will count as evidence for nesting, merger, factor removal, or a lower-dimensional alternative.

Pairwise Discriminating Contrasts

F1 Versus F2: Direct Boundary Adjudication

The canonical F1 manipulation - supplying an explicit state-action mapping - can also raise perceived efficacy or relative target-policy advantage, thereby changing F2. The framework therefore requires an explicit prospective F1/F2 adjudication rather than assigning an effect from the intervention label alone. Four outcomes are distinguished. (A) Support for separability: the specification manipulation measurably changes A while measured efficacy or P remains

adequately controlled or does not mediate the initiation effect; independent F2 manipulations change P without reproducing the proximal A pattern; and held-out perturbation responses favor a model containing both terms. (B) Mediation or nesting: action specification changes initiation primarily through a measured change in efficacy or P; F1 is then not independently identified by the supplied mapping, and the model should represent the relation as mediated or nested unless an additional F1 term improves prediction. (C) Merger or contraction: F1 and F2 perturbations become behaviorally interchangeable after proximal measurement and nuisance control, or a merged/lower-dimensional model predicts held-out responses equally well or better; F1 and F2 should merge for that task family. (D) Invalid test: the intended specification manipulation fails to change the prespecified A measure; the result does not count as negative evidence for F1.

Worked F1/F2 Contraction Case

Suppose, hypothetically, that supplying an explicit action mapping improves the prespecified A measure, but it also increases perceived efficacy and target-policy preference P. When P is modeled, the initiation advantage associated with the supplied mapping is carried by the change in P, and a model with a separate F1 term does not improve prediction for held-out perturbations. Under the framework's own rule, this is not “weak evidence for F1.” The appropriate result is mediation/nesting or merger for that task family: action specification has explanatory relevance through its effect on policy selection, but an additional independent F1 dimension has not earned retention.

F1 Versus F3: Action Construction Versus Operational Onset Support

One condition can require participants to generate the next valid action, whereas another supplies it. After A is measured and local target-policy selection P is observed, self-paced and externally supported onset can be compared. A double dissociation across independent

implementations would be stronger than a single rescue: specification manipulations should preferentially alter A, whereas onset-support manipulations should alter Y^I/T^I after the action is specified and $P = 1$. Because F3 lacks an independent proximal assay, this comparison can support a reproducible operational distinction but not by itself establish a unique enactment-coupling mechanism.

F2 Versus F3: Policy Selection Versus Operational Onset Support

Relative-value manipulations should occur before local policy choice and should show that P changed. Once $P = 1$ is observed, externally supported onset can test whether Y^I/T^I changes without a corresponding change in policy selection. A cueing effect after observed selection is evidence against a purely F2-level account, but F3 remains an operational onset contrast unless a non-circular proximal assay is developed; alerting, preparation, motor speed, expectancy, and convergent onset-support implementations remain necessary controls.

F3 Versus F4: Operational Onset Support Versus Transition Sensitivity

External onset support can reduce ambiguity about self-generated onset while repeat-versus-switch requirements are varied independently. A residual switch cost under matched external cueing is evidence for a transition-sensitive operational contrast not reducible to the specific self-generated-versus-supported onset contrast. It is not evidence for a unique transition mechanism. If adequately matched repeat and switch conditions become equivalent, F4 should be dropped for that context.

Illustrative Identification Protocol

A full empirical validation is beyond the scope of this theoretical article. The framework can, however, be instantiated in a learned finite-state control task in which participants move a system from a current state to a target state through deterministic actions. Such a task family

allows the same superordinate goal to be preserved while action specification, local policy value, onset support, and transition requirement are manipulated. The staged design is prospective: it is intended to let nested or lower-dimensional models win, not to guarantee recovery of four factors.

Table 2 separates manipulation assignment, proximal or validity evidence, the strict initiation measure, and the main alternative explanations that require control. It also adds an explicit F1/F2 boundary-adjudication step so that action specification cannot be treated as independent of policy value by assumption. These are proposed design elements, not validated assays.

Table 2

Illustrative staged protocol and inferential safeguards

Stage	Manipulation and proximal check	Strict initiation measure	Primary nuisance / alternative
1. Action construction	Self-generate first valid action vs supplied mapping; measure A (mapping accuracy/uncertainty) and concurrently measure efficacy/P.	First valid task-directed onset and T ⁱ after mapping becomes available.	Perceived efficacy, comprehension, instruction length, generalized facilitation, policy-value spillover.
2. Local policy selection	Manipulate relative payoff/efficacy/opportunity cost; verify target-choice probability or preference P and monitor A.	Whether $P = 1$; then Y^i/T^i where onset is subsequently analyzed.	Arousal, vigor, affect, conflict, demand characteristics, F1-mediated efficacy changes.
F1/F2 boundary adjudication	Independently vary specification and F2-directed perturbations where feasible; estimate A, efficacy/P, mediation/nesting, and compare F1+F2 against nested/merged models on held-out perturbations.	Y^i/T^i plus held-out perturbation prediction; no label assignment from intervention name alone.	Cross-target spillover, collinearity, measurement error, incomplete A or P engagement.

Stage	Manipulation and proximal check	Strict initiation measure	Primary nuisance / alternative
3. Operational onset support	After $P = 1$ and an executable action are observed, compare self-paced onset with external onset support; verify cue delivery/detection and match nonimperative alerting/preparation.	T^I and probability of onset within a prespecified window; endpoint response is not a proximal F3 assay.	Sensory alerting, temporal preparation, motor speed, expectancy; stronger F3 mechanism claim remains OPEN.
4. Transition sensitivity	With $P = 1$ and onset support matched, compare equally trained repeat vs switch requirements; verify task-set assignment.	Repeat-switch difference in T^I and onset error.	Conflict, remapping, working memory, unequal familiarity, preparation.
Downstream exclusion	After $Y^I = 1$, reveal low vs high execution demand where appropriate.	Not an initiation endpoint; classify post-onset failure separately as Z^E .	Learning and expectancy across repeated trials; motor capacity; task difficulty.

Worked End-to-End Attribution Sequence

A single hypothetical learned-task family illustrates how an investigator would use the framework. The sequence is not a reported study and does not presume that all four contrasts survive.

1. Establish G. Before manipulating immediate actions, confirm that the participant represents and endorses the target outcome for the episode. Permitted conclusion: $G = 1$. Forbidden conclusion: the target policy has been selected or onset is imminent. If the goal is not represented or endorsed, the episode falls outside the strict equivalence class and the initiation analysis stops.

2. Evaluate A. Compare self-generated versus supplied next-action specification and measure A independently of onset, while monitoring efficacy and P. Permitted conclusion: the manipulation changed action specification if A changes. Forbidden conclusion: an initiation rescue proves an independent F1 cause. If A does not change, the proposed F1 test is invalid; if its effect is carried by P, F1 nests or merges rather than surviving independently.

3. Evaluate P. With an executable action available, manipulate or measure relative target-policy advantage before onset. Permitted conclusion: the target policy is selected or has increased choice propensity when the prespecified P measure supports that statement. Forbidden conclusion: $P = 1$ guarantees initiation. Because measuring or forcing P can alter the later onset state, the design must address that reactivity rather than treating P conditioning as neutral.

4. Test the operational F3 contrast. After an executable action and $P = 1$ are established without manufacturing the onset effect, compare self-generated onset with appropriately controlled external onset support. A hypothetical cue rescue may show shorter T^I or more frequent onset, which supports a reproducible self-generated-versus-supported onset contrast. It does not identify a unique enactment mechanism. If the residual contrast does not survive valid convergent tests, F3 is dropped or merged.

5. Test transition sensitivity. With onset support approximately matched, compare equally trained repeat and switch requirements. A residual difference supports transition sensitivity at the operational level only. If ordinary switching, preparation, conflict, or related established constructs account for the effect, the model contracts to those constructs; if adequately matched repeat and switch conditions do not differ, F4 is dropped for that task family.

6. Classify downstream execution separately. Once the initiation criterion is crossed, measure Z^E under the planned execution demand. The pattern $Y^I = 1$ and $Z^E = 0$ is a downstream execution failure, not a second form of non-initiation. The sequence therefore ends with a scope decision as well as an attribution decision: initiation evidence cannot be back-filled from what happens after onset.

Cross-Task and Held-Out Validation

A function-specific interpretation requires more than one operationalization. Same-function contrasts should generalize across surface-different tasks better than they correlate with neighboring contrasts implemented by the same method. Held-out perturbations also matter: candidate models that fit the training manipulations but fail to predict new perturbation responses do not support identification. This requirement parallels the broader system-identification emphasis on informative perturbations and out-of-sample discrimination (Cipresso, 2026).

Model Contraction, Falsification, and Invalid Tests

Model contraction is a core success condition of the framework. The four labels are hypotheses about a useful decomposition, not entities that must survive. Future data can support separability, mediation or nesting, merger, dropping a contrast, a smaller/common-factor model, rejection of stable person-level profiles, or invalidation of a proposed selective test.

F1/F2 Boundary Adjudication

F1/F2 is the primary prospective boundary test. Evidence can support four outcomes: separability, mediation/nesting, merger, or an invalid test. Separability requires distinct proximal patterns in A and P plus superior held-out prediction from a model retaining both terms. Mediation/nesting is favored when specification affects initiation through measured efficacy or P and an additional F1 term adds no independent predictive value. Merger is favored when valid F1- and F2-directed perturbations are behaviorally interchangeable after proximal measurement or when a merged model predicts held-out responses as well as or better than the two-term model. The test is invalid when the F1 manipulation fails to engage A or the F2 manipulation fails to alter the prespecified policy-selection measure.

Merge Two Contrasts

If two perturbation families are validly implemented, obtain the prespecified proximal evidence where such evidence is available, and nevertheless produce interchangeable initiation effects across convergent implementations and nuisance controls, the corresponding contrasts should merge for that task family. For F1/F2, the explicit boundary rule above governs this decision. For F3/F4, which are deliberately broader operational contrasts, merger or removal depends on reproducible response patterns and held-out model performance rather than on a claimed unique mechanism assay.

Drop a Contrast

A proposed contrast should be dropped when valid tests show that it contributes no independent predictive or explanatory information beyond the remaining model. F3 should be removed or merged if no reproducible residual self-generated-versus-supported onset difference survives matched alerting/preparation controls and convergent implementations. F4 should be dropped for a task family if transition-sensitive costs disappear under adequately matched tests.

Prefer a Smaller or Common-Factor Model

Lower-dimensional alternatives must be compared directly, not relegated to a late caveat. A broad factor such as general control capacity, arousal, perceived efficacy, task difficulty, actionable expected control, or another common state may account for correlated sensitivity to several perturbations. Model comparison should therefore include F1/F2 nested and merged models, models without F3 or F4, and broader common-factor accounts, with measurement error represented where possible. If a smaller model predicts held-out perturbation responses as well as or better than the F1-F4 model, the smaller model should be preferred.

Reject a Stable Person-Level Profile

For individual i , a provisional four-component response vector may be written

$$R_i = (r_{1i}, r_{2i}, r_{3i}, r_{4i}),$$

but r_j initially denotes sensitivity to an operationalized contrast, not a demonstrated latent trait. A person-level profile requires cross-method and cross-occasion convergence. A useful directional criterion is

$$r(\text{same contrast across methods}) > r(\text{different contrasts within method}).$$

If method-specific variance dominates and same-contrast convergence is absent, stable person-level profiling should be rejected even if the endpoint remains underidentified at the episode level.

Invalid Selective Test

If a manipulation fails its prespecified validity criterion, or a required proximal variable such as A or P fails its prespecified engagement criterion, a null dissociation is not evidence that two contrasts are identical. The test is invalid for that inference. Conversely, once the relevant validity criteria and proximal evidence where defined are satisfied, repeatedly assigning every failed dissociation to unspecified off-target effects would make the framework non-adjudicating. Prespecified validity, proximal-evidence, and contraction rules are therefore necessary for falsifiability.

ADHD as an Application Domain

ADHD is a useful application because task initiation, reward sensitivity, effort allocation, delay, and state-dependent performance are prominent in its literature. The framework does not define ADHD, and diagnosis does not determine a unique F1-F4 profile.

In particular, the framework does not assert ADHD = universally elevated initiation threshold or ADHD = weak motivation. If recruitment or admission thresholds are introduced in later empirical models, their effective levels should be treated as individual- and state-dependent parameters unless evidence supports a stronger group claim. Cue weighting, transition sensitivity, or reliance on external structure likewise should not be assumed from diagnosis alone.

Cai and Mizuno (2026) review dynamic neurocognitive approaches that move beyond a single inhibitory-control or reward account. Katabi and Shahar (2024), using hierarchical reinforcement-learning models, reported group differences particularly apparent for initiatory actions rather than uniformly across an action sequence. Winter et al. (2019) found similar rates of high-effort choice in children with ADHD and controls but lower successful execution in the ADHD group.

The Winter et al. result is used here only to reinforce the downstream exclusion: choice and post-initiation execution can dissociate, so execution failure should not be folded into a strict initiation endpoint. These findings do not establish the F1-F4 factorization. The empirical direction remains: measure operational contrasts first, then test whether ADHD phenotype predicts their distribution or dynamics.

Biological localization is OPEN. No one-to-one mapping from F1-F4 to a neurotransmitter system, brain region, developmental pathway, or diagnostic subtype is claimed.

History, State, and Intervention Labels

The four contrasts should not be mistaken for stable person types. A more plausible representation is

$$Y_{i,t}^l = F(\theta_i, z_{i,t}, X_t),$$

where θ_i denotes relatively persistent history-bearing parameters, $z_{i,t}$ current state, and X_t current task and environmental conditions. The same person can therefore show different limiting patterns across tasks or occasions.

History is treated as a parameter source rather than as an additional function. Learning and prior experience may alter reward weighting, expected efficacy, social-evaluative cost, sensitivity to urgency, or the effectiveness of external onset support. The framework does not infer which parameter changed merely because an environmental manipulation helped.

Common intervention labels are therefore not mechanisms. “Body doubling,” task splitting, deadlines, reward, and accountability are packages that can perturb several variables at once. Task splitting may reduce action-construction demand while increasing efficacy; a deadline may increase opportunity cost while also increasing threat; another person may provide temporal cueing, alter competition, or change social value. Mechanistic claims require decomposition of the intervention package, not its everyday label.

Implications for Motivation and Action-Control Research

Non-Initiation Should Not Be Treated as a Direct Measure of One Latent Cause

The practical implication is not merely that task initiation has many influences. It is that the same observed failure can remain compatible with different candidate organizations even after a superordinate goal has been endorsed. Treating $Y^1 = 0$ as a direct measure of weak motivation, deficient intention, or any other single latent cause therefore collapses a measurement problem into a diagnosis. Motivation research can instead preserve the failure endpoint while asking which additional observations and perturbations are needed to discriminate among candidate explanations.

Intervention Response Requires an Identification Standard

A task begins after an implementation intention, an external cue, added reward, a deadline, or another person becomes present; such a rescue is informative, but it does not identify its own mechanism. Intervention packages can simultaneously alter action specification, efficacy, relative policy value, alerting, preparation, social evaluation, and competition. For F1 and F2, independently defined proximal measures can support stronger attribution; for F3 and F4, the present framework supports operational contrasts more securely than unique latent-mechanism claims. The useful unit of evidence is therefore the joint pattern of manipulation validity, proximal change where measurable, initiation response, convergent discrimination, and held-out prediction.

For motivation theory, contraction is itself substantive evidence. If an apparent action-specification effect is fully mediated by local policy preference, the result favors a more economical motivational account of that task family rather than a four-part decomposition. Likewise, a cue rescue can establish a useful onset-support dependency without licensing a new latent mechanism. The framework is intended to change the interpretation of familiar goal-to-action failures by specifying when a motivational explanation has been earned, when it remains ambiguous, and when it should be simplified.

Initiation Is Not Execution, Persistence, or Termination

High competence once a task is active is compatible with poor entry into that task, and strong persistence after entry does not imply easy initiation. Conversely, normal initiation can coexist with downstream execution failure. Motivation research often spans all of these phenomena, but they should not be collapsed into one behavioral endpoint. The present

framework analyzes initiation only and uses post-initiation execution solely as an exclusion contrast.

What the Framework Does and Does Not Establish

The article specifies a bounded conceptual problem: under the cited motivation and action-control literatures, strict non-initiation can be compatible with more than one candidate functional organization. It proposes an F1–F4 task-initiation hypothesis class and an ordered sequence of perturbations, proximal measurements, conditioning rules, and contraction criteria for evaluating whether that added structure is useful.

It does not establish that four biologically discrete modules or orthogonal latent factors exist, that the four contrasts are exhaustive, that each failed episode has one unique limiting function, that intervention labels identify mechanisms, that F3 has an independently validated proximal assay, that the illustrative protocol already achieves target-specific identification, that stable person-level profiles exist, or that ADHD is defined by any one threshold or mechanism.

The article also makes no novelty claim for the intention-behavior gap, general Action Control Theory, implementation intentions, value- and efficacy-sensitive control allocation, controlled perturbation, target-engagement logic, or behavioral system identification. The residual contribution is the task-initiation-specific candidate organization, its explicit F1/F2 boundary adjudication, the ordered ambiguity-reduction protocol, and the prospective contraction logic that refuses to diagnose mechanism from the failure endpoint alone.

Limitations and Open Scientific Questions

First, F1-F4 may still be overcomplete or incomplete. The most immediate empirical question is whether F1 and F2 remain separable once action specification, efficacy, and local policy selection are measured together. A mediated, nested, or merged F1/F2 model is an

admissible outcome. Prospective memory, uncertainty resolution, hierarchical planning, social coordination, persistence, and termination may also require additional distinctions in other task families.

Second, selective manipulation is difficult. Explicit instructions can change efficacy; reward can change arousal and vigor; external cues can change alerting and preparation; and repeat-versus-switch manipulations recruit multiple processes. The framework therefore requires convergent perturbations and clearly separated manipulation-validity, proximal-evidence, and endpoint measures rather than assuming selectivity by design.

Third, progressive ambiguity reduction does not fix latent states. Experimental ordering can reduce known ambiguities, but residual cross-stage interactions, learning, and expectancy remain possible and must be controlled or modeled.

Fourth, stable individual profiles are OPEN. Repeated, cross-method measurement is required before treating response parameters as history-bearing person characteristics.

Fifth, biological interpretation is OPEN. The cited neurological and ADHD findings constrain possible interpretations but do not localize F1-F4 to unique biological substrates.

Sixth, the proposed finite-state protocol is illustrative. Its F1/F2 proximal measures, mediation or nesting models, manipulation-validity criteria, thresholds for invalid tests, and held-out comparison procedures require empirical development. F3 remains explicitly operational unless a non-circular proximal assay is developed. Future results may support the proposed distinctions, force contraction, or favor a different candidate factorization.

Conclusion

Task-initiation failure is an observable endpoint, not a diagnosis of its cause. Under the baseline condition $G = 1$, non-initiation can remain compatible with several candidate functional

organizations. F1 and F2 concern distinctions before or during local target-policy selection; F3 and F4 are onset contrasts evaluated only after $P = 1$ has been observed or made sufficiently stable. Failure after onset is a different endpoint and remains outside the strict initiation factorization.

The proposed contribution to motivation science is therefore an identification discipline for the goal-to-action boundary. It combines a task-initiation-specific candidate F1–F4 organization with explicit conditioning, direct F1/F2 adjudication, evidence-matched treatment of F3 and F4, ordered perturbations, and prospective rules for nesting, merger, dropping, common-factor alternatives, invalid tests, and rejection of stable person profiles. The framework is designed so that fewer distinctions can win.

The resulting rule is not “observe non-initiation and infer motivation,” nor simply “perturb and infer.” It is to define the endpoint and conditioning variables; distinguish superordinate goal endorsement from local policy selection and onset; validate the manipulation; obtain independent proximal evidence where possible; measure initiation; test neighboring and lower-dimensional alternatives; and retain added dimensions only when they improve held-out prediction. Whether four contrasts, a nested F1/F2 relation, fewer contrasts, or a different decomposition ultimately survives remains an empirical question.

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